evolves into an entangled superposition of the different possible measurement outcomes.

But there's an alternative possibility. Before the measurement happened, there was one electron and one observer (or camera, if you prefer—it doesn't matter how we think about the thing that interacts with the electron as long as it's a big, macroscopic object). After they interact, however, rather than thinking of that one observer having evolved into a superposition of possible states, we could think of them as having evolved into *multiple possible observers*. The right way to describe things after the measurement, in this view, is not as one person with multiple ideas about where the electron was seen, but as *multiple worlds*, each of which contains a single person with a very definite idea about where the electron was seen.

Here's the big reveal: what we've described as austere quantum mechanics is more commonly known as the Everett, or Many-Worlds, formulation of quantum mechanics, first put forward by Hugh Everett in 1957. The Everett view arises from a fundamental annoyance with all of the special rules about measurements that are presented as part of the standard textbook quantum recipe, and suggests instead that there is just a single kind of quantum evolution. The price we pay for this vastly increased elegance of theoretical formalism is that the theory describes many copies of what we think of as "the universe," each slightly different, but each truly real in some sense. Whether the benefit is worth the cost is an issue about which people disagree. (It is.)

In stumbling upon the Many-Worlds formulation, at no point did we take ordinary quantum mechanics and tack on a bunch of universes. The potential for such universes was always there—the universe has a wave function, which can very naturally describe superpositions of many different ways things could be, including superpositions of the whole universe. All we did is to point out that this potential is naturally actualized in the course of ordinary quantum evolution. Once you admit that an electron can be in a superposition of different locations, it follows that a person can be in a superposition of having seen the electron in different locations, and indeed that reality as a whole can be in a superposition, and it becomes natural to treat every term in that superposition as a separate "world." We didn't add anything to quantum mechanics, we just faced up to what was there all along.